HYPER-LOCAL AVIATION WEATHER FORECAST VERIFICATION SYSTEM FOR SRI LANKAN MONSOON PATTERNS

30°C

MAX. 31°C - MIN.15°C

20%

15 KM/H

WEST

1023.0

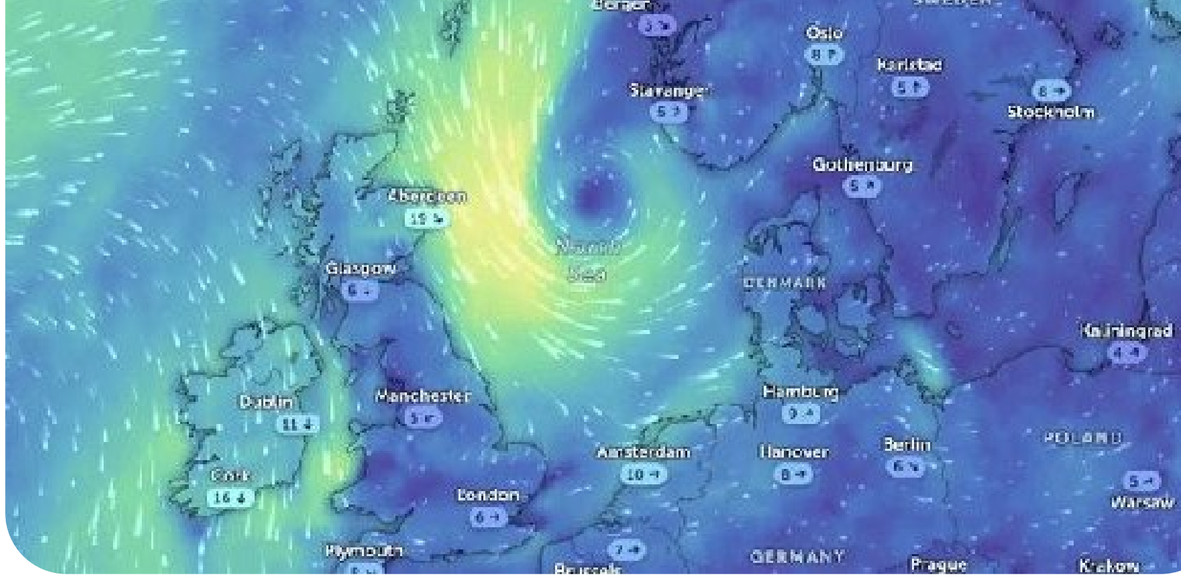
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BACKGROUND

Weather forecasting plays a critical role in aviation safety and operational decision-making. Pilots depend on aviation weather reports such as TAF and METAR to obtain atmospheric conditions before take-off and landing



PROBLEM

Most aviation weather forecasting systems rely on global models such as GFS and ECMWF, which provide large-scale forecasts but often miss local airport weather variations.

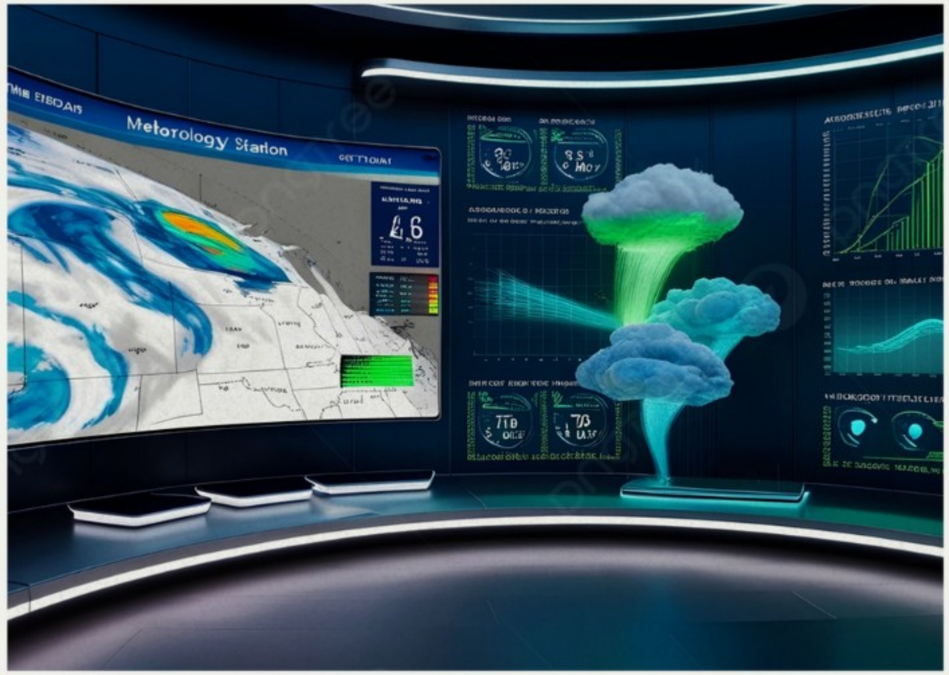
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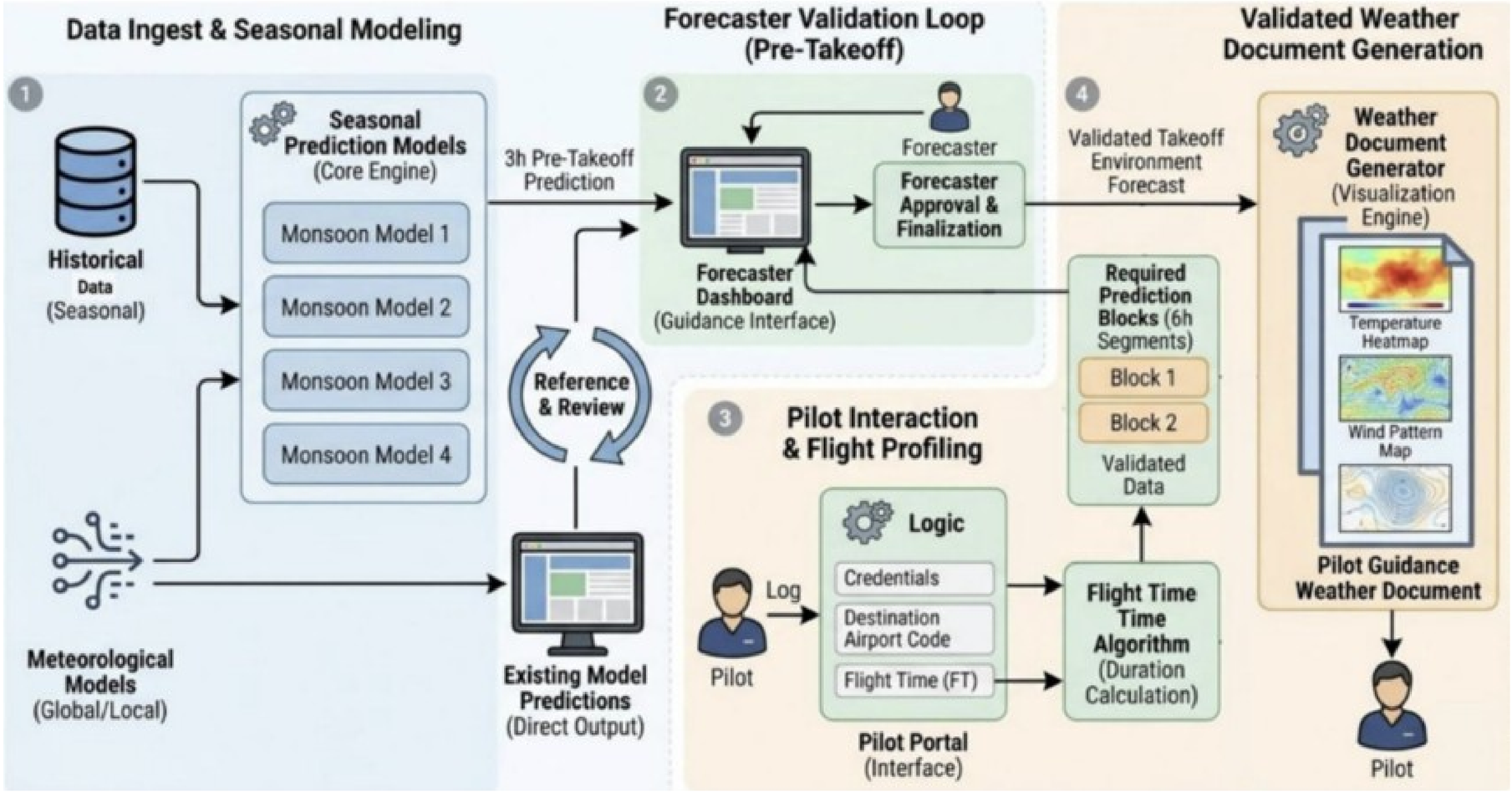
20% 15 KM/H WEST 1023.0

THE PROPOSED SOLUTION

- **Microclimate aviation weather prediction system**
- **Predict the next 3 hours weather da**
- **This system works as an aviation weather verification system**
- **Provides automated weather verification**
  support for forecasters and pilots



SYSTEM ARCHITECTURE



SANKADEEPA W.D.I

IT22113986



Pressure-Temperature Coupled Prediction Model

NORTH EAST MONSOON

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PRESSURE-TEMPERATURE COUPLED PREDICTION MODEL

Sub Problem

Existing large-scale forecasting systems are Fail for Larage scale prediction for temperature and pressure during North-East Monsoon conditions at BIA airport.

Solutions

- Uses AI-based coupled prediction for 3-hour temperature and pressure forecasting.
- Uses hyper-local microclimate aviation weather data.
- Integrates METAR and TAF verification support.
- Supports short-term aviation weather forecasting.

User Requirements

- Accurate 3-hour weather prediction
- Reliable forecast verification
- Short-term aviation weather monitoring
- Better operational decision-making

Design Excellence / Contribution

- Applied hyperparameter tuning to improve prediction accuracy
- Implemented a coupled prediction mechanism for temperature and pressure
- Performed feature engineering using lag-based and rolling statistical features
- Integrated METAR and TAF based forecast verification support

PRESSURE-TEMPERATURE COUPLED PREDICTION MODEL

User Feedback on Prototype



- The prototype improved understanding of temperature and pressure trend analysis during North-East Monsoon conditions.
- Users suggested integrating live aviation weather data in future versions.
- System improved view of Temp and pressure trend analysis
- Prototype demonstrated practical support for forecast verification

Technologies

- Python
- Scikit-Learn
- TensorFlow/Keras
- Random Forest
- LSTM
- Pandas & NumPy

Ranasinghe R.A.D.
IT22269720



Wind Pattern Prediction Model for Southwest Monsoon

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Wind Pattern Prediction Model (speed and direction)

Sub Problem

Global models fail to detect rapid wind speed and direction changes at Bandaranaike International Airport, affecting aviation safety under changing climate conditions.

Solution

- train ML models to predict short-term wind direction and speed using Micro Climate data
- Converts predicted wind data into headwind and crosswind components.
- Provides runway-oriented aviation weather guidance.

User Requirements

- short-term wind speed and wind direction prediction
- Runway-oriented headwind and crosswind calculetion
- Forecast verification using METAR and TAF aviation weather reports
- Real-time monitoring of rapidly changing wind conditions
- create automating weather report generation and delivery process

Design Excellence

- implement the different model using the different algorithms.
- Applied hyperparameter tuning to improve wind prediction accuracy and model performance (using GridSearchCV/RandomizedSearchCV)
- use formulas for calculete wind prediction mechanism for crosswind and headwind analysis
- Develop an automatic model retraining pipeline

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Ranasinghe

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Wind Pattern Prediction Model (speed and direction)

User Feedback on Prototype

- requested clearer visualization of wind speed ,crosswind and hedwind individual colums
- Users suggested integrating live aviation weather data APi.
- suggested creating an automated pipeline to retrain ML model monthly
- Prototype demonstrated practical support for forecast verification
- suggested using the last 3-6h data as input for the prediction models.

Technologies Used

- Python
- Scikit-Learn (random forest)
- Random Forest
- LSTM
- Pandas & NumPy
- mstplotlib

PERFORMANCE BENCHMARK METRICS

Forecast Type	Model Name	Accuracy (R^2 Score)	Error (Mean Absolute Error - MAE)
Short-Term Forecast (1-Hour Nowcast)	XGBoost Model (Production)	**67.38%**	**1.31 Knots**
Medium-Term Forecast (3-Hour Future Prediction)	XGBoost Model (Production)	**49.39%**	**1.69 Knots**
Medium-Term Forecast (3-Hour Future Prediction)	LSTM (Deep Learning) Model	**47.89%**	**1.74 Knots**
Medium-Term Forecast (3-Hour Future Prediction)	Random Forest Model	**47.84%**	**1.73 Knots**

Jayawardhana H.M.S.I

IT22619976



Cloud and Visibility Prediction Model for aviation Safety

First Inter-Monsoon period

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CLOUD AND VISIBILITY PREDICTION MODEL FOR AVIATION SAFETY-FIRST INTER MONSOON PERIOD.

Solution Addresses the Sub-Problem / Prototype

Sub Problem

Rapid visibility and cloud changes during the
First Inter-Monsoon period
are difficult to predict accurately at BIA airport.

User Requirements

- Accurate visibility and cloud prediction
- Real-time weather monitoring support
- Safer take-off operations at BIA
- Better local weather forecasting

Solutions

- Uses models to predict visibility and cloud conditions.
- Analyzes METAR weather data to identify rapid weather changes.
- Helps pilots and make safer take-off decisions.
- Provides early warnings for low visibility conditions.
- Improves local airport weather prediction accuracy.

Design Excellence / Contribution

- Compared Random Forest, XGBoost, and Decision Tree models
- Used weather factors like RH%, Temperature, and QNH
- Applied automated data cleaning and preprocessing
- Improved prediction accuracy using AI techniques

CLOUD AND VISIBILITY PREDICTION MODEL FOR AVIATION SAFETY-FIRST INTER MONSOON PERIOD.



Algorithm	R² Score	Performance Level
Random Forest Regressor	51%	Best Performing
XGBoost Regressor	48%	Moderate
Decision Tree	40%	Poor

User Feedback on Prototype

- The system helped identify sudden weather changes
- It improved visibility and cloud analysis
- Users found the AI verification useful
- Suggested adding live METAR data in future versions

Technologies

- Python
- Scikit-Learn
- XGBoost
- Random Forest
- Pandas & NumPy

Jayathilaka P.K.V.K

IT22069504



Relative Humidity & QNH Stability Prediction Model

Second Inter-Monsoon

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RELATIVE HUMIDITY & QNH STABILITY PREDICTION MODEL FOR AVIATION SAFETY-SECOND INTER MONSOON PERIOD

Solution Addresses the Sub-Problem / Prototype

Sub Problem

Relative Humidity and QNH changes during the Second Inter-Monsoon period are difficult to predict accurately at BIA airport.

User Requirements

- Accurate RH and QNH prediction
- Real-time 3-Hour Predictions
- Safer take-off operations at BIA
- Better local weather forecasting
- Improved decision-making for pilots and airport authorities

Solutions

- Uses AI models to predict QNH and RH conditions.
- Analyzes METAR weather data to identify rapid weather changes.
- Helps pilots and make safer take-off decisions.
- Provides early warnings for low visibility conditions.
- Improves local airport weather prediction accuracy.

Design Excellence / Contribution

- Compared Random Forest, XGBoost, and LightGBM
- Feature Engineering to create a custom Stability Index. By calculating the ratio between humidity and pressure
- Applied automated data cleaning and preprocessing
- Improved prediction accuracy using AI techniques

RELATIVE HUMIDITY & QNH STABILITY PREDICTION MODEL FOR AVIATION SAFETY-SECOND INTER MONSOON PERIOD.

Model	R² Score / Accuracy	Performance Level
LightGBM	83%	Best Performing
XGBoost	79%	Moderate
Random Forest	73%	Poor

User Feedback on Prototype

- Users found the prediction dashboard easy to understand
- Forecast outputs were useful for aviation weather monitoring
- System improved visibility of weather trend analysis
- Prototype demonstrated practical support for forecast verification

Technologies & AI Models Used

- Python
- Scikit-Learn
- TensorFlow/Keras
- Random Forest
- LSTM
- Pandas & NumPy

- https://epicflightacademy.com/metar/
- https://skybrary.aero/articles/meteorological-aerodrome-report-metar
- https://meteo.gov.lk
- Federal Aviation Administration (2023). Pilot’s Handbook of Aeronautical Knowledge (FAA-H-8083-25C). Chapter 11: Aircraft Performance. U.S. Department of Transportation.
- Key Focus: Mathematical derivation of crosswinds and trigonometric applications.
  Link: https://www.faa.gov/sites/faa.gov/files/13_phak_ch11.pdf

Hyper-Local Aviation Weather Forecast Verification System for Sri Lankan Monsoon Patterns

Commercialization & Sustainability

Customer Persona

- Primary: Airport Authorities (e.g., AASL) & Air Traffic Control (ATC) units.
- Secondary: Airline Flight Operations Departments (Dispatchers).
- End-Users: Commercial Pilots (Decision Support) and Meteorological Forecasters.e.

Unique Selling Point (USP)

- Built for Sri Lanka’s Main Airport (BIA
- Uses real weather physics (pressure, temperature, wind)
- Not just prediction — understands weather relationships
- Human-in-the-loop system for expert verification

Market Size

- Global smart airport market: ~$25 billion by 2030
- High demand in South Asia (Sri Lanka, India, Maldives)
- Focus: Airports affected by monsoon weather delays
- BIA handles 40,000+ flights per year

Cost & Revenue Model

- Subscription model for airports (SaaS)
- Saves airlines fuel cost ($2,000–$5,000 per avoided delay/diversion)
- Premium API for flight planning systems

Thank You

